

ABARES Working Paper

Published: 22 December 2025

Authors: Neal Hughes, Maruge Zhao, Andrew John, Avril Horne

Introduction

The Australian Murray-Darling Basin is home to one of the world's most mature water markets. In recent decades, these markets have played a vital role in supporting adaptation to the effects of drought, climate change and environmental reform. This study introduces a new monthly economic model of the water market with an emphasis on biophysical detail and empirical performance.

This model also contains significant economic structure, representing the stochastic and dynamic nature of water markets in storage-controlled rivers, including forward looking water users with rational expectations over future conditions. In this study, the model is applied to simulate the potential effects of climate change on water markets within the southern basin. The results show how markets support adaptation, with drier climates leading to increases in regional water trade volumes, particularly imports into the lower Murray.

Drier scenarios are also associated with more conservative crop planting decisions, which increase storage reserves to maintain supply reliability. Under the driest future climates, the ability of markets to adapt is more limited, and the model simulates long-term declines in irrigation development. However, the future climate remains highly uncertain, with a majority of the projection ensemble involving increases in water supply and irrigation activity relative to recent conditions.

In future, fully integrated hydro-economic models could help design water market institutions that are robust to a non-stationary and uncertain climate.

Source: <https://www.agriculture.gov.au/abares/research-topics/working-papers/water-markets-under-climate-change-murray-darling-basin-model>